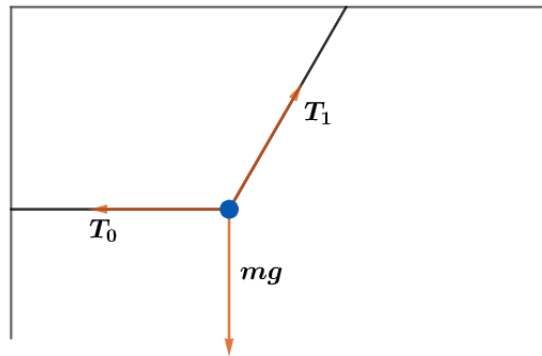


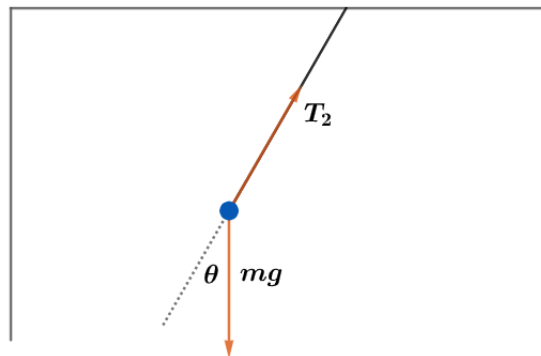
2023 F=ma Exam: Problem 12

Kevin S. Huang



Initially, we balance forces to obtain

$$T_1 = \sqrt{(mg)^2 + T_0^2} > mg$$



After the horizontal string is cut, the tension in the diagonal string balances the radial component of gravity,

$$T_2 = mg \cos \theta < mg$$

Thus, we have $T_2 < mg < T_1$ so the answer is B.